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**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY**

2022 INVITATIONAL B

Biology

- B01. C
- B02. C
- B03. A
- B04. B
- B05. D
- B06. E
- B07. E
- B08. B
- B09. C
- B10. D
- B11. A
- B12. C
- B13. B
- B14. B
- B15. A
- B16. D
- B17. E
- B18. D
- B19. E
- B20. A

Chemistry

- C01. A
- C02. B
- C03. C
- C04. D
- C05. A
- C06. D
- C07. C
- C08. E
- C09. B
- C10. A
- C11. E
- C12. E
- C13. B
- C14. B
- C15. A
- C16. C
- C17. B
- C18. A
- C19. C
- C20. B

Physics

- P01. C
- P02. B
- P03. D
- P04. E
- P05. B
- P06. D
- P07. A
- P08. C
- P09. D
- P10. C
- P11. A
- P12. B
- P13. E
- P14. A
- P15. B
- P16. D
- P17. C
- P18. B
- P19. A
- P20. E

CHEMISTRY SOLUTIONS – UIL INVITATIONAL B 2022

- C01. (A) The 4d section of the periodic table includes elements 39-48.
- C02. (B) $\text{NaBrO}_2 = 22.99 + 79.90 + 16.00 \times 2 = 134.89 \text{ g/mol} \times \frac{1}{2} \text{ mol} = 67.45 \text{ g}.$
- C03. (C) You have $0.088 \text{ L} \times 0.052 \text{ mol/L} = 0.004576 \text{ mol Pb}^{2+}$ so you can form $0.00458 \text{ mol PbI}_2$. $0.00458 \text{ mol PbI}_2 \times 461.0 \text{ g/mol} = 2.1 \text{ g}$
- C04. (D) Wavelength and energy are inversely proportional, so a longer wavelength in a photon means a lower energy.
- C05. (A) Phosphorus trichloride has four regions of electron density around the central phosphorus atom, giving it tetrahedral electronic geometry. Since one of those regions of electron density is a lone pair, the molecular shape is trigonal pyramidal.
- C06. (D) The containers have the same temperature and the same number of moles, so the change in pressure depends only on the change in volume. $P_A V_A = P_B V_B$, so $P_B = P_A (V_A/V_B)$. Since their diameters are the same, the volume ratio is equal to the height ratio, and $V_A/V_B = 300/400$. $P_B = 1.50(300/400) = 1.125 \text{ atm}.$
- C07. (C) Formaldehyde is a polar molecule, so its strongest intermolecular forces are dipole-dipole forces.
- C08. (E) The metal oxidation states are A) +1; B) +2, C) +4, D) +5, E) +3
- C09. (B) The water that the chemicals are dissolved in is part of the surroundings, and the surroundings gained heat, which means the system gave off heat, making the reaction exothermic.
- C10. (A) The heat of fusion for ice is 334 J/g and you have 454 grams , so $334 \times 454 = 151,636 \text{ J}$. To heat the water from 0°C to 25°C , $q = mc\Delta T = (454 \text{ g})(4.184 \text{ J/g}^\circ\text{C})(25^\circ\text{C}) = 47,488 \text{ J}$. The total energy required is $199,124 \text{ J} = 199 \text{ kJ}$.
- C11. (E) The equilibrium expression is product concentrations raised to their stoichiometric coefficients divided by reactant concentrations raised to their stoichiometric coefficients, but you leave out the solids and liquids.
- C12. (E) KOH and Ba(OH)_2 are both strong bases, but on a mole-to-mole basis Ba(OH)_2 releases twice as many hydroxide ions as KOH and will have a higher pH (by 0.3 pH units) at the same concentration. BaCl_2 and KCl are both neutral salts, and H_2SO_4 is a strong acid and would have the lowest pH below 7. (Barium hydroxide is not as soluble as KOH , but at this concentration all of the Ba(OH)_2 dissolves.)
- C13. (B) For a 1:1 salt like XY , $K_{sp} = [x][y] = x^2$, where x is the molar solubility of the salt. The molar solubility therefore equals $\sqrt{K_{sp}} = (6.2 \times 10^{-5})^{1/2} = 7.9 \times 10^{-3}$.
- C14. (B) $E^\circ_{\text{cell}} = E^\circ_{\text{reduction}} - E^\circ_{\text{oxidation}}$. We are not told which electrode is the anode and which is the cathode, but we are told that it is an electrolytic cell, which means the cell potential is negative. (In a galvanic cell the half-reaction with the higher standard potential is the cathode; in an electrolytic cell the half-reaction with the lower standard potential is the cathode.) $E^\circ_{\text{red}} = -0.76 \text{ V}$ and $E^\circ_{\text{ox}} = +0.80 \text{ V}$. $E^\circ_{\text{cell}} = (-0.76 \text{ V}) - (+0.80 \text{ V}) = -1.56 \text{ V}$

- C15. (A) Except for F^- these are all isoelectronic (same electron configuration), so the one with the fewest protons in the isoelectronic series would be the largest. The fluoride ion will be smaller than Cl^- so we don't need to know where it falls among the cations in order to answer the question. The actual radii in pm are Cl^- (167) > K^+ (152) > F^- (119) > Ca^{2+} (114) > Ga^{3+} (76) [Wikipedia].
- C16. (C) An intermediate is produced as a product early in the mechanism and consumed as a reactant in a later step, so it does not appear in the overall chemical equation for the reaction.
- C17. (B) Solve for the total volume of the gas at 1 atm and $25^\circ C$, then divide by the volume of one balloon to determine how many balloons can be filled. $PV = nRT$, so $V = nRT/P$. $n = 12.8$ mol, $T = 298$ K, $R = 0.08206$ L·atm/mol·K, and $P = 1.00$ atm. $V = (12.8)(0.08206)(298)/1.00 = 313$ L total volume. Each balloon is 6.25 L, so the number of balloons is $313 \text{ L} \div 6.25 \text{ L/balloon} = 50.08$ balloons.
- C18. (A) $(8.50 \times 10^{25} \text{ atoms}) / (6.02 \times 10^{23} \text{ atoms/mole}) = 141.2$ moles of Fe. $141.2 \text{ moles} \times 55.85 \text{ g/mol} = 7885.8 \text{ grams} = 7.89 \text{ kg Fe}$
- C19. (C) Boiling point elevation and freezing point depression are two colligative properties of solutions.
- C20. (B) $100 \text{ g CaCl}_2 / 110.98 \text{ g/mol} = 0.90106$ moles $CaCl_2 \times 2 \text{ mol } Cl^- / \text{mol } CaCl_2 = 1.802 \text{ mol } Cl^-$.
 $1.802 \text{ mol } Cl^- / 0.500 \text{ L} = 3.60 \text{ M}$

PHYSICS SOLUTIONS – UIL INVITATIONAL B 2022

- P01. (C) page 20: “But because Faraday had no formal education, he did not have the command of the mathematics that would allow him to describe his remarkable discoveries. Instead, he filled up notebooks with strange diagrams showing lines of force that look like the web of lines iron filings make when surrounding a magnet. He also invented the concept of a field, one of the most important concepts in all of physics.”
- P02. (B) page 45: “Any theory, no matter how beautiful, must eventually confront experimental verification. So, Einstein seized upon several possible experiments. The first was the erratic orbit of Mercury.”
- P03. (D) pages 65-66: “If the electron had wavelike properties, then what was disturbing the medium in which the wave existed? What was waving?... According to a paper by Max Born in 1926, what was waving was the probability of finding an electron at that point.... In other words, electrons are particles, but the probability of finding the particle at any given location is given by a wave function.”
- P04. (E) Long period comets have orbital periods greater than 200 years, and their orbits are often highly inclined to the ecliptic. This suggests that these comets originate in the spherical shell of ice bodies known as the Oort Cloud, which extends from 2000 to 200,000 AU from the Sun.
- P05. (B) When using any analog measurement tool, like a ruler, you should always estimate the measurement to one decimal place better than the smallest markings on the tool. In this case, the smallest markings are the millimeter scale, so we should estimate the measurement to the nearest tenth of a millimeter.
To complete this specific measurement, we must read the ruler at both ends of the metal bar. One end, to the nearest tenth of a millimeter is at 1.45cm. The other end is at 7.88cm. The length of the bar is the difference between these two readings: $L = 7.88 - 1.45 = 6.43\text{cm}$.
- P06. (D) The units are odd, but they are compatible, so we don't need to make any unit conversions to solve the problem. We have distance information but are not given any information about the time that the car is on the entrance ramp. Thus, we should use the kinematic equation: $v_f^2 = v_i^2 + 2a(x_f - x_i)$.
Plugging in the given values: $v_f^2 = (30.0)^2 + 2(9500)(0.15) = 3750$. So $v_f = \sqrt{3750} = 61\text{mph}$.
- P07. (A) Since the incline is frictionless, the forces acting on the crate are as follows: gravity, directed downwards; the normal force, directed up and left, perpendicular to the plane; and the applied force, directed up and right, parallel to the plane. As is typical with an inclined plane, we rotate our coordinate system so that the x-axis is parallel to the plane and the y-axis is perpendicular to the plane. Thus, the applied force is in the positive x-direction and the normal force is in the positive y-direction. The gravitational force must be broken into components: in the negative x-direction is the component $mg \sin \theta$ and in the negative y-direction is the component $mg \cos \theta$. There is no motion in the y-direction, so we will ignore it and move on to the x-direction. Adding up the forces in the x-direction gives: $\sum F_x = F - mg \sin \theta = ma = 200.0 - (50.0)(9.8) \sin 22 = (50.0)a$ This gives: $(50.0)a = 200 - 183.6 = 16.4 \rightarrow a = 0.329\text{m/s}^2$. To find the velocity, we will use the kinematic equation: $v_f = v_i + at = 0 + (0.329)(2.50) = 0.822\text{m/s}$.
- P08. (C) This problem requires us to use conservation of momentum. Taking “to the right” to be positive, the initial momentum is: $p_i = m_1 v_{1i} + m_2 v_{2i} = (0.150)(1.50) + (0.220)(0) = 0.225 \text{ kgm/s}$. And the final momentum is given by:
 $p_f = m_1 v_{1f} + m_2 v_{2f} = (0.150)(-0.250) + (0.220)(v_{2f}) = -0.0375 + 0.220v_{2f}$.
By conservation of momentum, the final momentum must equal the initial momentum, thus:
 $p_i = p_f \rightarrow 0.225 = -0.0375 + 0.220v_{2f} \rightarrow 0.220v_{2f} = 0.2625 \rightarrow v_{2f} = 1.19\text{m/s}$.

- P09. (D) If the system is balanced, that means that the sum of the forces add up to zero and the sum of the torques also add up to zero. Summing up forces is not helpful since we do not know the magnitude of the upward Normal Force provided by the fulcrum. Instead, we will solve the problem by summing up torques.

First, we must choose a pivot point. For a lever system like this, it makes sense to choose the pivot point to be at the fulcrum. All torque-arm distances will be measured from this pivot point. Torque is defined at the product of the force, the torque-arm distance, and the sine of the angle between the torque-arm and the force. Fortunately for us, the forces in this problem are all perpendicular to the torque-arm (the torque arm is parallel to the meter stick). Thus, the sine of those 90-degree angles is equal to 1 and does not affect the calculations. Therefore, we need the product of the forces and the distances of those forces from the fulcrum. We must do some conversions as well – masses into kilograms, distances into meters, and we have to multiply the hanging masses by g .

For the known masses, we get torques of $\tau_1 = (0.1000\text{kg})(9.8)(0.600\text{m} - 0.300\text{m}) = 0.294\text{Nm}$ and $\tau_2 = (0.0500\text{kg})(9.8)(0.800\text{m} - 0.300\text{m}) = 0.245\text{Nm}$. The meter stick itself has a mass (effectively acting at the center-of-mass), so it also produces a torque:

$$\tau_3 = (0.0400\text{kg})(9.8)(0.500\text{m} - 0.300\text{m}) = 0.0784\text{Nm}.$$

The Normal Force at the fulcrum provides no torques since its torque-arm length is zero.

All of these torques (so far) are trying to make the meter stick rotate clockwise. Thus, they will all be negative when we sum up the torques. The only torque that is trying to make the meter stick rotate counterclockwise is that produced by the unknown mass. This counterbalancing torque, which will be positive, is $\tau_4 = (M)(9.8)(0.300\text{m} - 0.0\text{m}) = 2.94M$.

Now we can sum up all the torques and set the sum to zero. Taking into account their signs we get:

$$\sum \tau = -0.294 - 0.245 - 0.0784 + 2.94M = 0.$$

$$\text{This gives: } 2.94M = 0.6174 \rightarrow M = 0.2100\text{kg} = 210.0\text{g}$$

- P10. (C) Since the sound wave expands spherically, the absolute intensity is the power divided by the surface area of the spherical wavefront, given by $I = \frac{P}{4\pi r^2}$. In this problem that gives the result:

$$I = \frac{13.0\text{ W}}{4\pi(4.20\text{m})^2} = 0.0586 \frac{\text{W}}{\text{m}^2}.$$

To convert this into decibels, we use $I(\text{dB}) = 10\text{Log}\left(\frac{I}{I_0}\right)$, where $I_0 = 10^{-12} \frac{\text{W}}{\text{m}^2}$. Thus, we get $I(\text{dB}) = 10\text{Log}\left(\frac{0.0586}{10^{-12}}\right) = 10(10.8) = 108\text{ dB}$.

- P11. (A) Since the system is floating, we know that the downward gravitational force is exactly balanced by the upward buoyant force. The total downward gravitational force includes both the weight of the rock and the weight of the disk: $F_g = M_{\text{rock}}g + M_{\text{disk}}g$. Since we want the density of the disk, we will rewrite its mass in terms of its density: $M_{\text{disk}} = \rho_{\text{disk}}V_{\text{disk}}$. This gives a total gravitational force of $F_g = M_{\text{rock}}g + \rho_{\text{disk}}V_{\text{disk}}g$.

The buoyant force is based on the density of the liquid (pure water) and the volume of the disk submerged under the surface. Since 50% of the disk is above the surface, we know that 50% of the disk must be below the surface. Thus, the buoyant force is $F_B = \rho_{\text{water}}V_{\text{under}}g = \rho_{\text{water}}\frac{1}{2}V_{\text{disk}}g$.

Setting the gravitational force equal to the buoyant force and putting in our given and known values (in mks units):

$$F_g = M_{\text{rock}}g + \rho_{\text{disk}}V_{\text{disk}}g = F_B = \rho_{\text{water}}\frac{1}{2}V_{\text{disk}}g$$

$$(12.0)(9.8) + \rho_{\text{disk}}(0.157)(9.8) = (1000)\left(\frac{1}{2}\right)(0.157)(9.8)$$

$$117.6 + 1.5386\rho_{\text{disk}} = 769.3 \rightarrow 1.5386\rho_{\text{disk}} = 651.7 \rightarrow \rho_{\text{disk}} = 424\text{kg/m}^3.$$

- P12. (B) Since the capacitor is initially uncharged, we should use the RC charging equations. The equation for the amount of charge stored on a capacitor during the charging phase is given by $Q = CV(1 - e^{-t/RC})$. All of these quantities, including the time, are given.
 Thus, $Q = (320 \times 10^{-6})(18.0)(1 - e^{-(3)/[(6500)(320 \times 10^{-6})]})$.
 This simplifies to $Q = (5.76 \times 10^{-3})(1 - e^{-1.442}) = (5760 \mu C)(0.7635) = 4400 \mu C$.
- P13. (E) First, we need to find the electric field between the plates. For parallel plates, the electric field is uniform and constant, and given by $E = \frac{V}{d}$. Converting the distance into meters, this gives an electric field of $E = \frac{15.0V}{0.0750m} = 200V/m$. From this we can determine the electric force on the particle of dust:
 $F = qE = (30.0 \times 10^{-6})(200) = 6.00 \times 10^{-3}N$. We also know the mass of the piece of dust, so we can find its acceleration using Newton's second law. Converting the mass into kilograms, we get:
 $F = ma = 6.00 \times 10^{-3} = (12.0 \times 10^{-6}kg)(a)$. This gives the acceleration:
 $a = \frac{6.00 \times 10^{-3}}{12.0 \times 10^{-6}} = 500 m/s^2$. Finally, we can use kinematics to determine the velocity of the piece of dust when it reaches the negative plate:
 $v_f^2 = v_i^2 + 2a(x_f - x_i) = 0 + 2(500)(0.0400) = 40 \rightarrow v_f = \sqrt{40} = 6.32m/s$.
- P14. (A) This is an application of Faraday's Law, which states that the voltage induced in a conductor by a changing magnetic flux is given by $\mathcal{E} = -\frac{d\Phi}{dt} \approx -\frac{\Delta\Phi}{\Delta t}$. Here the magnetic flux is $\Phi = BA$, the magnetic field strength multiplied by the area of the loop. We don't need to worry about an angle since the field is perpendicular to the face of the loop. The loop is square, so the area of the loop is:
 $A = \ell^2 = (0.18)^2 = 0.0324m^2$. The area is constant, so $\mathcal{E} = -\frac{\Delta\Phi}{\Delta t} = -\frac{\Delta(BA)}{\Delta t} = -A\frac{\Delta B}{\Delta t} = -A\frac{B_f - B_i}{\Delta t}$.
 Converting the magnetic field to Tesla, and bringing it all together:
 $\mathcal{E} = -A\frac{B_f - B_i}{\Delta t} = -(0.0324)\frac{0.0430 - 0.0070}{9.00 \times 10^{-3}} = -(0.0324)\frac{0.0360}{0.009} = -0.1296V$. So, $|\mathcal{E}| = 0.1296V$.
 Lastly, we turn to Ohm's Law to find the current: $I = \frac{\mathcal{E}}{R} = \frac{0.1296}{1.25} = 0.104A = 104mA$.
- P15. (B) In order to determine the speed of light in a substance, you need the index of refraction of that substance. Fortunately, we can use the angles given, and Snell's law, to get the index of refraction. Snell's law states $n_1 \sin \theta_1 = n_2 \sin \theta_2$, and since the light starts in air, we know $n_1 = 1.00$. Plugging this, and the angles, into the equation, we get:
 $(1.00) \sin(40.0) = n_2 \sin(20.3) \rightarrow 0.6428 = 0.3469n_2$. So, the index of refraction of the crystal is $n_2 = 1.85$. The speed of light in a substance equals the speed of light in a vacuum divided by the index of refraction. Thus, the speed of light in the crystal is
 $v = \frac{c}{n_2} = \frac{3.00 \times 10^8}{1.85} = 1.62 \times 10^8 m/s$.
- P16. (D) The information that your image is 1/10 of your normal height gives the magnification of the optical system. This relates to the object (p) and image (q) locations: $M = -\frac{q}{p} = 0.100 = -\frac{q}{1.80}$. Thus, the image location is $q = -0.180m$. That it is negative indicates that it is a virtual image.

 To determine the radius of curvature, we utilize the mirror equation to find the focal length:
 $\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \rightarrow \frac{1}{1.80} + \frac{1}{-0.180} = \frac{1}{f} \rightarrow f = -0.200m = -20.0cm$. The radius of curvature of a mirror is twice the focal length of the mirror: $R = 2f = 2(-20.0) = -40.0cm$. Thus, the magnitude of the radius of curvature of the polished metal sphere is: $|R| = |-40.0| = 40.0cm$.

- P17. (C) First, we need the energy of the photons shining on the metal surface. Using the equation for the energy of a photon, we get: $E_\gamma = \frac{hc}{\lambda} = \frac{1240 \text{ eVnm}}{460 \text{ nm}} = 2.70 \text{ eV}$. Now we must subtract the workfunction of the metal to find the energy of the emitted electrons: $E_e = E_\gamma - \phi = 2.70 - 1.60 = 1.10 \text{ eV}$. This is well below relativistic energies, so we can equate this to the classical kinetic energy of the electrons: $\frac{1}{2}mv^2 = 1.10 \text{ eV}$. Converting the energy units and putting in the mass of an electron gives us $(1.10 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV}) = (0.5)(9.11 \times 10^{-31} \text{ kg})v^2 \rightarrow v^2 = 3.86 \times 10^{11}$. This gives an emitted electron velocity of $v = 6.21 \times 10^5 \text{ m/s}$.
- P18. (B) First, we should notice that the particle contains a quark-antiquark pair. This means that the particle is a meson (baryons contain three quarks). Thus, the answer is neither D nor E. Now, notice that there are no strange quarks/antiquarks in the particle, this means that the strangeness must be zero. Thus, C cannot be the correct answer. Finally, we know that an up quark has a charge of $+\frac{2}{3}e$, but a charm antiquark has a charge of $-\frac{2}{3}e$. Combined, these charges cancel – leaving the particle with a net charge of zero. Thus, A is not the correct answer. Only answer choice B correctly describes the properties of this particle. By the way, this particle is cataloged as the $\bar{D}^0$ meson.
- P19. (A) Thermal expansion is approximately linear and is described by the equation $\Delta L = \alpha L_0 \Delta T$. The data collected form a straight line; and, based on the equation, the slope of the line would be equal to $\text{slope} = \frac{\Delta L}{\Delta T} = \alpha L_0$. For the slope, I used two points that are reasonably far apart: $(20^\circ\text{C}, 2.115 \text{ m})$ and $(100^\circ\text{C}, 2.141 \text{ m})$. From these, I get a slope of: $\text{slope} = \frac{2.141 - 2.115}{100 - 20} = \frac{0.026 \text{ m}}{80^\circ\text{C}} = 3.25 \times 10^{-4} \frac{\text{m}}{^\circ\text{C}} = \alpha L_0$. For L_0 we use the length at 60°C (the temperature of interest) and calculate the coefficient of thermal expansion to be: $3.25 \times 10^{-4} \frac{\text{m}}{^\circ\text{C}} = \alpha(2.128 \text{ m}) \rightarrow \alpha = 1.53 \times 10^{-4} ^\circ\text{C}^{-1} \approx 1.5 \times 10^{-4} ^\circ\text{C}^{-1}$.
- P20. (E) The power of a lens is the inverse of the focal length of the lens, with the units of diopters being equal to $(\text{meters})^{-1}$. Nearsightedness means that the far point of the patient's eye is too small. A normal far point should be very large – we usually use a far point of infinity to represent normal vision. Therefore, to correct nearsightedness, we need a lens that takes an object at infinity and produces an image at the patient's far point: negative 33.0cm in this case. The image location must be negative because the image should be out in front of the person's face – on the same side of the corrective lens as the object. Using the lens equation: $\frac{1}{p} + \frac{1}{q} = \frac{1}{f} = \text{Power}$ with $p = \infty$ and $q = -0.33 \text{ m}$, we can get the power of the corrective lens needed: $\frac{1}{\infty} + \frac{1}{-0.33} = \text{Power} = 0 - 3.0 = -3.0 \text{ diopter}$. This is the power of the lens needed to correct this patient's nearsightedness.